Problem :

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Key questions

1. Are correct input types and attributes used?

Yes. Correct HTML5 input types and attributes like text, email, tel, password, radio, checkbox, id, and placeholder are used appropriately.

2. Are validation errors displayed near the relevant fields?

No. Currently, validation errors are displayed using alert pop-ups, not directly near the relevant fields.

3. Is password strength checked?

Yes. Password strength is checked using Regex for a minimum of 8 characters, including at least one letter and one number.

4. Is the form accessible using keyboard and screen-reader-friendly Labels? Yes. The form can be accessed using the keyboard, and labels are provided for the input fields, making it screen-reader-friendly.

Code:

1)index .html

```
<!DOCTYPE html>
<html>
<head>
    <title>Registration Form</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
  <h2>Registration Form</h2>
  <form id="registrationForm">
   <label for="name">Name:</label>
  <input type="text" id="name" placeholder="Enter your name">
  <span id="nameError" class="error"></span>
 <label for="email">Email:</label>
 <input type="text" id="email" placeholder="example@gmail.com">
 <span id="emailError" class="error"></span>
```

```
<label for="phone">Phone Number:</label>
 <input type="text" id="phone" placeholder="10 digit number">
 <span id="phoneError" class="error"></span>
 <label for="password">Password:</label>
 <input type="password" id="password" placeholder="Enter password">
 <span id="passwordError" class="error"></span>
 <label for="confirmPassword">Confirm Password:</label>
 <input type="password" id="confirmPassword"
placeholder="Re-enter password">
<span id="confirmPasswordError" class="error"></span>

 <label for="course">Course:</label>
  <select id="course">
 <option value="">Select Course</option>
 <option value="BCA">BCA</option>
 <option value="B.Tech">B.Tech</option>
 <option value="MCA">MCA</option>
  <option value="M.Tech">M.Tech</option>
 </select>

       <span id="courseError" class="error"></span>
 <label for="year">Year:</label>
 <select id="year">
 <option value="">Select Year</option>
 <option value="1">1st Year</option>
 <option value="2">2nd Year</option>
  <option value="3">3rd Year</option>
 <option value="4">4th Year</option>
       </select>
       <span id="yearError" class="error"></span>
         <label>Gender:</label>
<div class="gender">
           <input type="radio" name="gender" value="Male">
           Male
<input type="radio" name="gender" value="Female">
           Female
<input type="radio" name="gender" value="Other">
           Other
       </div>
```

```
  <span id="genderError" class="error"></span>
         <div class="terms">
  <input type="checkbox" id="terms">
 <label for="terms">
                I accept Terms and Conditions
            </label>
        </div>
        <span id="termsError" class="error"></span>
        <button type="submit">Register</button>
  <p id="successMessage" class="success"></p>
    </form>
</div>
<script src="script.js"></script>

</body>
</html>
```

2)style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}

.container {
    width: 400px;
    margin: 40px auto;
    padding: 25px;
  background-color: white;
  border-radius: 10px;
box-shadow: 0 0 10px gray;
}

h2 {
    text-align: center;
}
label {
    display: block;
    margin-top: 12px;

    font-weight: bold;
```

```
}
input[type="text"],
input[type="password"],
select {
    width: 100%;
  padding: 10px;
     margin-top: 5px;
 box-sizing: border-box;
 border: 1px solid gray;
 border-radius: 5px;
}
.gender {
    margin-top: 8px;
}
.gender input {
    margin-left: 10px;
}

.terms {
    margin-top: 15px;
}
.terms input {
    margin-right: 5px;
}

.error {
    display: block;
color: red;
  font-size: 13px;
 margin-top: 3px;
}
.success {
color: green;
 text-align: center;
 font-weight: bold;
}
button {
    width: 100%;
 padding: 10px;
  margin-top: 20px;
```

```
 background-color: blue;
  color: white;
    border: none;
      border-radius: 5px;
 cursor: pointer;
}
button:hover {
    background-color: darkblue;
}
```

3)script.js

```
const form = document.getElementById("registrationForm");
form.addEventListener("submit", function(event) {
    event.preventDefault();
    let name =
        document.getElementById("name").value.trim();

    let email =
        document.getElementById("email").value.trim();

    let phone =
        document.getElementById("phone").value.trim();

    let password =
        document.getElementById("password").value;

    let confirmPassword =
        document.getElementById("confirmPassword").value;

    let course =
        document.getElementById("course").value;

    let year =
        document.getElementById("year").value;

    let terms =
        document.getElementById("terms").checked;
    document.getElementById("nameError").innerHTML = "";
document.getElementById("emailError").innerHTML = "";
 document.getElementById("phoneError").innerHTML = "";
```

```
 document.getElementById("passwordError").innerHTML = "";
 document.getElementById("confirmPasswordError").innerHTML = "";
 document.getElementById("courseError").innerHTML = "";
 document.getElementById("yearError").innerHTML = "";
 document.getElementById("genderError").innerHTML = "";
 document.getElementById("termsError").innerHTML = "";
 document.getElementById("successMessage").innerHTML = "";
 let valid = true;
    let namePattern = /^[A-Za-z ]+$/;
 if (name === "") {
  document.getElementById("nameError").innerHTML =
"Please enter your name.";
   valid = false;
  }
  else if (!namePattern.test(name)) {
 document.getElementById("nameError").innerHTML =
     "Name should contain only letters.";
  valid = false;
    }
    let emailPattern =
   /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
 if (email === "") {
  document.getElementById("emailError").innerHTML =
   "Please enter your email.";
  valid = false;
  }
  else if (!emailPattern.test(email)) {
 document.getElementById("emailError").innerHTML =
  "Please enter a valid email.";
    valid = false;
    }
    let phonePattern = /^[0-9]{10}$/;
    if (phone === "") {
   document.getElementById("phoneError").innerHTML =
  "Please enter phone number.";
  valid = false;
  }
 else if (!phonePattern.test(phone)) {
  document.getElementById("phoneError").innerHTML =
   "Phone number must be 10 digits.";
```

```
 valid = false;
   }
 let passwordPattern =
       /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d).{8,}$/;
   if (password === "") {
 document.getElementById("passwordError").innerHTML =
           "Please enter password.";
valid = false;
}
else if (!passwordPattern.test(password)) {
  document.getElementById("passwordError").innerHTML =
"Password must contain 8 characters, uppercase, lowercase and number.";
  valid = false;
   }

   if (confirmPassword === "") {
document.getElementById("confirmPasswordError").innerHTML =
           "Please confirm password.";
valid = false;

   }
else if (password !== confirmPassword) {
document.getElementById("confirmPasswordError").innerHTML =
           "Passwords do not match.";
 valid = false;
   }

   if (course === "") {

       document.getElementById("courseError").innerHTML =
           "Please select course.";
  valid = false;
   }
if (year === "") {
document.getElementById("yearError").innerHTML =
           "Please select year.";
 valid = false;
   }
   let gender =
document.querySelector(
```

```
            'input[name="gender"]:checked'
        );
    if (!gender) {
 document.getElementById("genderError").innerHTML =
            "Please select gender.";
  valid = false;
    }
 if (!terms) {
 document.getElementById("termsError").innerHTML =
  "Please accept Terms and Conditions.";
   valid = false;
    }
    if (valid) {
 document.getElementById("successMessage").innerHTML =
            "Registration successful!";

        form.reset();
    }

});
```

Index1.html

```
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Student Hub Home</title>

    <link rel="stylesheet" href="../style.css">
    <link rel="stylesheet" href="../css/about.css">
    <link rel="stylesheet" href="../css/assignment.css">
    <link rel="stylesheet" href="../css/attendence.css">
    <link rel="stylesheet" href="../css/contact.css">
    <link rel="stylesheet" href="../css/course.css">
    <link rel="stylesheet" href="../css/login.css">
    <link rel="stylesheet" href="../css/profile.css">
    <link rel="stylesheet" href="../css/result.css">
```
```

```
    <link rel="stylesheet" href="../css/studentfeedback.css">
</head>

<body>

    <header>
        <h2>Student Hub</h2>
    </header>

    <img src="../images/charusat.png"
         alt="CHARUSAT University Logo"
         width="10%"
         height="60">

    <nav>
        <a href="index.html">Dashboard</a>
        <a href="login.html">Login</a>
        <a href="profile.html">Profile</a>
        <a href="about.html">About</a>
        <a href="assignment.html">Assignment</a>
        <a href="attendance.html">Attendance</a>
        <a href="contact.html">Contact</a>
        <a href="course.html">Course</a>
        <a href="result.html">Result</a>
        <a href="studentsfeedback.html">Student Feedback</a>
    </nav>

    <main>
        <h1>Welcome to Student Hub Portal</h1>

        <p>
            This portal provides information about assignments,
            attendance, contact, courses, dashboard, login,
            registration, results and student feedback.
        </p>

        <hr>

        <h2>Regex Validation Demo</h2>
```

```
        <form id="validationForm" novalidate>

            <label for="email">Email Address:</label>
            <br>

            <input
                type="text"
                id="email"
                name="email"
                placeholder="Enter your email"
                required
            >

            <br><br>

            <label for="password">Password:</label>
            <br>

            <input
                type="password"
                id="password"
                name="password"
                placeholder="Enter your password"
                required
            >

            <br><br>

            <button type="submit">Validate</button>
            <button type="reset">Reset</button>

            <p id="message" role="status" aria-live="polite"></p>

        </form>
    </main>

    <footer>
        <h6>
            &copy; 2026 CHARUSAT University | Made by ID No. 25DCE023
        </h6>
```

```
    </footer>

    <script src="../js/about.js"></script>
    <script src="../js/assignment.js"></script>
    <script src="../js/attendence.js"></script>
    <script src="../js/contact.js"></script>
    <script src="../js/course.js"></script>
    <script src="../js/login.js"></script>
    <script src="../js/profile.js"></script>
    <script src="../js/result.js"></script>
    <script src="../js/studentfeedback.js"></script>

    <script>
        // Regex pattern for email validation
        const emailRegex =
            /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

        // Regex pattern for password validation
        // Minimum 8 characters, uppercase, lowercase,
        // number and special character
        const passwordRegex =

/^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;

        const form = document.getElementById("validationForm");
        const emailInput = document.getElementById("email");
        const passwordInput = document.getElementById("password");
        const message = document.getElementById("message");

        form.addEventListener("submit", function (event) {

            // Prevent page refresh
            event.preventDefault();

            const email = emailInput.value.trim();
            const password = passwordInput.value;

            // Check empty fields
            if (email === "" || password === "") {
                message.textContent = "Please fill in all fields.";
```

```
                return;
            }

            // Validate email using Regex
            if (!emailRegex.test(email)) {
                message.textContent =
                    "Invalid email! Please enter a valid email address.";
                return;
            }

            // Validate password using Regex
            if (!passwordRegex.test(password)) {
                message.textContent =
                    "Password must contain at least 8 characters, " +
                    "one uppercase letter, one lowercase letter, " +
                    "one number and one special character.";
                return;
            }

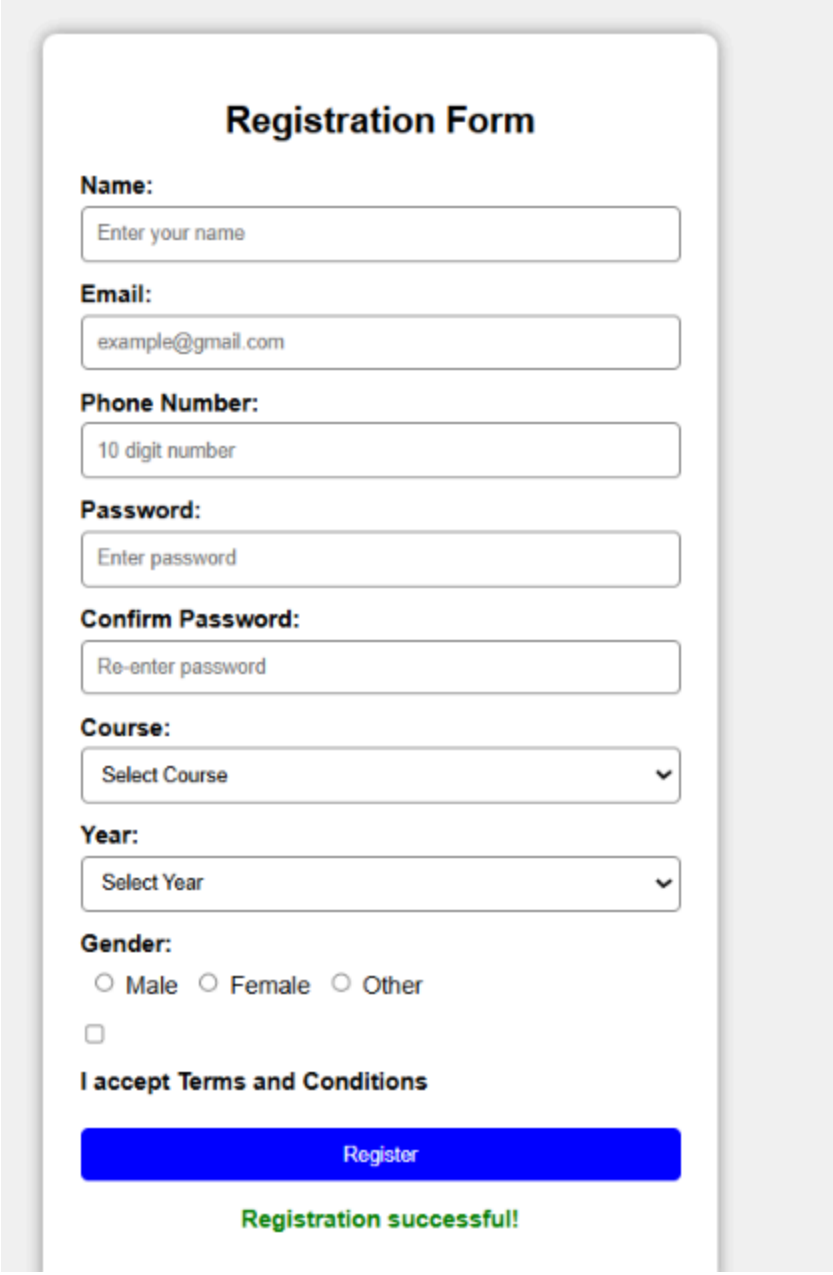
            // If both validations pass
            message.textContent =
                "Email and password format validated successfully!";
        });

        // Clear validation message on reset
        form.addEventListener("reset", function () {
            message.textContent = "";
        });
    </script>

</body>
</html>
```

Output:


Student Hub
Dashboard
Login
Profile
About
Assignment
Attendance
Contact
Course
Result
Student Feedback
Welcome to Student Hub Portal
This portal provides information about assignments, attendance, contact, courses, dashboard, login, registration, results and student feedback.
Regex Validation Demo
Email Address:
nency1709@gmail.com
Password:
Validate
Reset
Email and password format validated successfully!
© 2026 CHARUSAT University | Made by ID No. 25DCE023


Registration Form
Name:
Enter your name
Email:
example@gmail.com
Phone Number:
10 digit number
Password:
Enter password
Confirm Password:
Re-enter password
Course:
Select Course
Year:
Select Year
Gender:
Male
Female
Other
I accept Terms and Conditions
Register
Registration successful!